

EW – 1 Lab 5

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INTRODUCTION TO SOLDERING:

Soldering is the process of using a wire of solder alloy to fix metal joints in a circuit permanently. There are many pencil style irons suitable for lab use, but we used a 25W iron because that's the bare minimum.

Soldering flux is used to fix the joints more efficiently, otherwise cold-joints are potential.

AIM:

- Design and solder a simple RC circuit.
- Design and solder a simple voltage regulator circuit using a Zener diode.
- Compare the readings of the soldered circuits with previous labs.

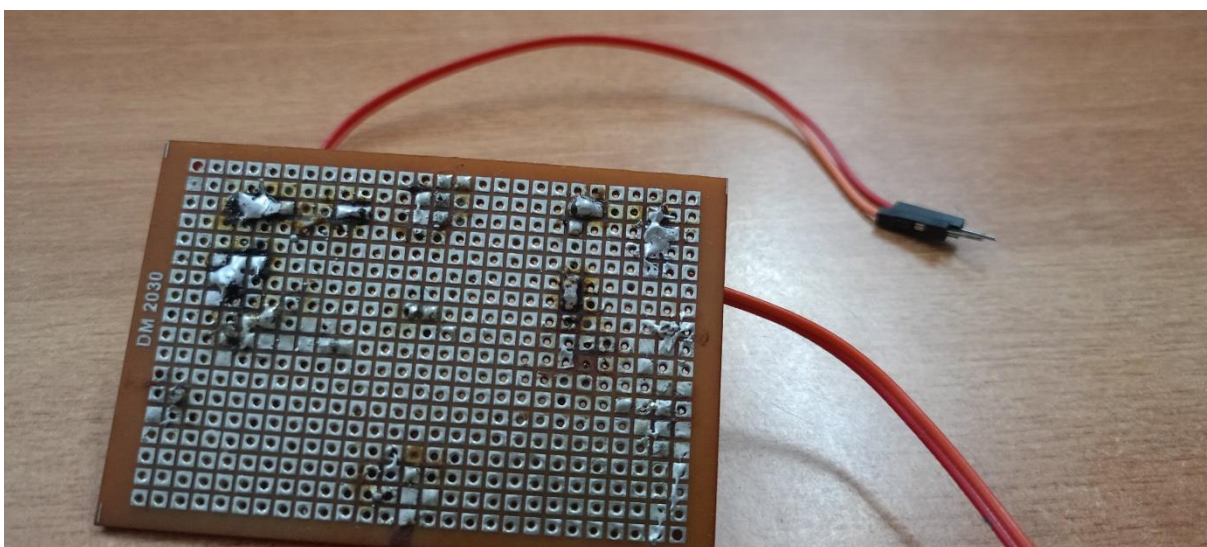
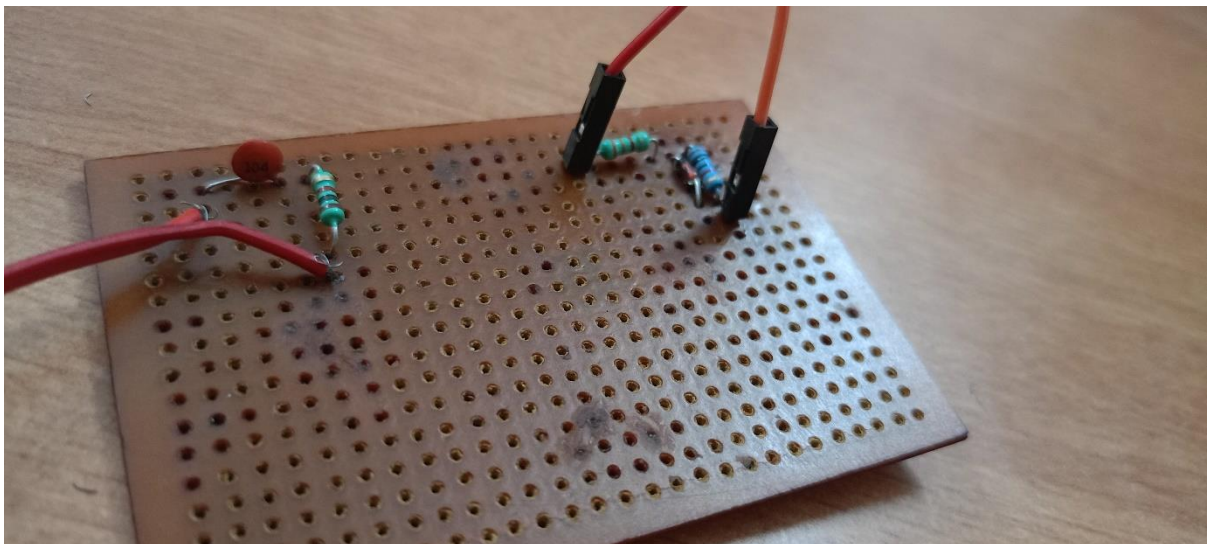
Components Required:

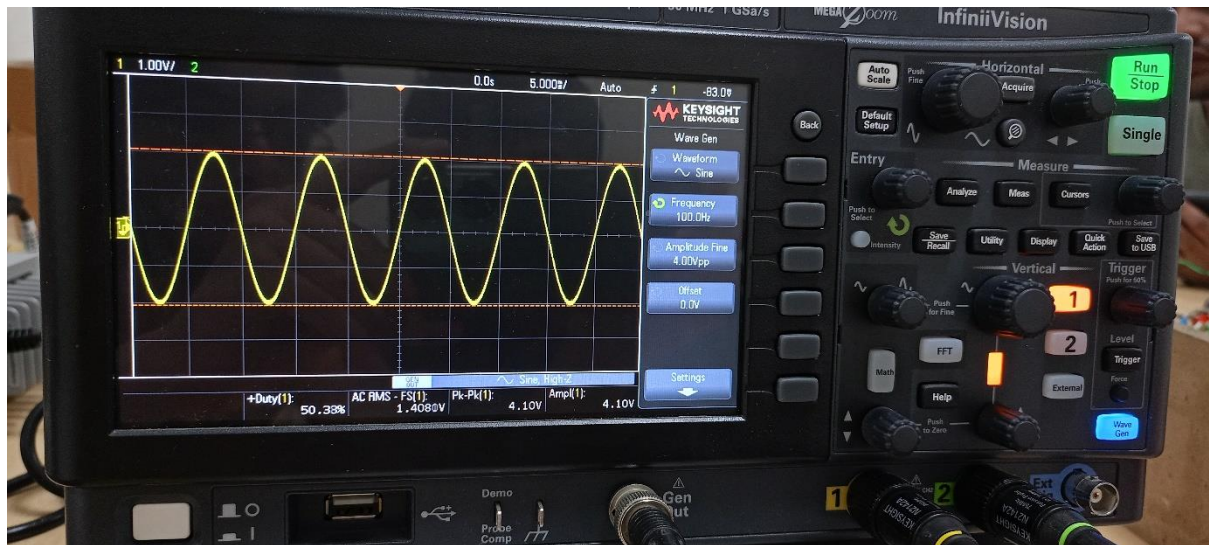
- Solder pad
- Soldering iron
- Solder flux
- Solder lead
- Wires
- Resistors
- Capacitors
- Zener Diode (5.1V)

Procedure:

- Place the components for RC circuit on the solder pad.
- Apply flux and heat the joints, then solder them.
- Repeat the same for the Zener circuit.
- Note readings for both the circuits (input: 15V) and compare with previous results.

Results:





Observations: The readings of the circuits matched with the previous circuits.

Conclusion: The voltage regulator and RC circuits were successfully soldered.